

Albin P Shaji

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EDUCATION

TKM College Of Engineering

Kollam, Kerala

Bachelor of Technology in Computer Science and Engineering | CGPA: 7.84

May 2023 – May 2027 (Expected)

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), C, Python, SQL (PostgreSQL, MySQL), HTML/CSS

Frameworks & Libraries: Spring Boot, React.js, Node.js, Axios, JUnit, Lombok

Infrastructure & Tools: Docker, Git, Linux, Postman, VS Code

Core Competencies: RESTful API Design, Relational Database Design, Containerization

PROJECTS

Sevana - AI Powered Donation Website | Java, Spring Boot, React.js, PostgreSQL, PostGIS, AI/ML, DockerLIVE

- Architected a full-stack **Circular Economy Marketplace** to optimize resource distribution between donors and NGOs, utilizing **Spring Security** and **JWTs** for strict Role-Based Access Control.
- Integrated an **AI-powered chatbot** and smart **AI-powered recommendation engine** to automate user support and intelligently match donor surplus with specific NGO resource deficits.
- Engineered a geospatial logistics dashboard utilizing **PostGIS** and interactive maps, empowering NGOs to visualize real-time donor locations and calculate **optimized multi-stop pickup routes**.
- Implemented a gamified Donor Leaderboard utilizing complex **PostgreSQL aggregations** to track and rank contributions, significantly driving **user engagement**.
- Utilized Docker to containerize the Spring Boot backend, isolating application dependencies and streamlining deployment readiness for multi-platform cloud hosting.

Generative AI Document RAG Engine | Next.js, TypeScript, PostgreSQL (pgvector), Vercel AI SDK, LLM APIs, NLP

- Built a local Retrieval-Augmented Generation document question-answering application using Next.js, TypeScript, PostgreSQL, pgvector, and Ollama.
- Implemented a text ingestion pipeline to normalize, chunk, embed, and store document content as 768-dimensional vector embeddings using nomic-embed-text.
- Configured PostgreSQL with the pgvector extension and HNSW cosine similarity indexing to enable semantic search over stored document chunks.
- Integrated a local LLM inference pipeline using Ollama and qwen3.5:4b, exposing API routes for document ingestion, vector retrieval, and grounded answer generation.

P2P Rental Marketplace (Rentopiaa) | React.js, Node.js, Express.js, MongoDB, Socket.io

LIVE

- Architected and developed a full-stack **peer-to-peer rental marketplace** using the **MERN stack**, enabling users to seamlessly list, discover, and rent products.
- Engineered a robust **rental booking engine** with advanced **date-range overlap validation** algorithms, preventing concurrency issues like double-bookings and ensuring fair availability.
- Integrated a secure payment gateway using the **Razorpay API** to process complex marketplace transactions, including dynamic rental fees and automated **refundable security deposits**.
- Built a real-time messaging system utilizing **Socket.IO**, facilitating instant, low-latency communication and negotiation between product owners and renters.

Expense Tracker | Java, JavaFX, MySQL, JDBC

Jan 2024

- Built an **MVC-compliant** desktop application utilizing **JavaFX** for the graphical interface and **JDBC** for robust, persistent database transactions.
- Programmed strict data isolation protocols at the database layer, establishing a secure **multi-tenant environment** that prevents cross-user data leakage.
- Engineered a **dynamic financial dashboard** that parses **SQL queries** into visual data representations, allowing users to track and analyze monthly expenditure trends efficiently.

COURSEWORK

Data Structures and Algorithms, Operating Systems, Database Management Systems (DBMS), Object-Oriented Programming (OOP), Computer Networks, Software Engineering